Xinyang Geng

1780 Spruce St.
Apt 13
Berkeley, CA 94709

young.geng@berkeley.edu
(510) 612-4352
http://young-geng.xyz

EDUCATION

University of California, Berkeley
Ph.D. in Computer Science — Aug 2018 – Now

- Advised by Professor Sergey Levine.
- Research in language models, vision-language models and deep reinforcement learning.

University of California, Berkeley
B.A. in Computer Science and Statistics (double major) — Aug 2013 – May 2017

- Graduated with high distinction and honor in computer science.

PUBLICATIONS AND PREPRINTS

Offline Q-Learning on Diverse Multi-Task Data Both Scales And Generalizes
Aviral Kumar*, Rishabh Agarwal*, **Xinyang Geng**, George Tucker, Sergey Levine
Notable top 5%, ICLR 2023, arXiv:2211.15144

Towards Better Few-Shot and Finetuning Performance with Forgetful Causal Language Models
Hao Liu*, **Xinyang Geng***, Lisa Lee, Igor Mordatch, Sergey Levine, Sharan Narang, Pieter Abbeel
arXiv:2210.13432

Multimodal Masked Autoencoders Learn Transferable Representations
Xinyang Geng*, Hao Liu*, Lisa Lee, Dale Schuurams, Sergey Levine, Pieter Abbeel
arXiv:2205.14204

Design-bench: Benchmarks for data-driven offline model-based optimization
Brandon Trabucco*, **Xinyang Geng***, Aviral Kumar, Sergey Levine
Oral presentation, ICML 2022, arXiv:2202.08450

Conservative Objective Models for Effective Offline Model-Based Optimization
Brandon Trabucco*, Aviral Kumar*, **Xinyang Geng**, Sergey Levine
Oral presentation, ICML 2021, arXiv:2107.06882

Variable-Shot Adaptation for Online Meta-Learning
Tianhe Yu*, **Xinyang Geng***, Chelsea Finn, Sergey Levine
arXiv:2012.07769

Meta-Reinforcement Learning Robust to Distributional Shift via Model Identification and Experience Relabeling
Russell Mendonca*, **Xinyang Geng***, Chelsea Finn, Sergey Levine
arXiv:2006.07178

Rewriting History with Inverse RL: Hindsight Inference for Policy Improvement
Benjamin Eysenbach*, **Xinyang Geng***, Sergey Levine, Ruslan Salakhutdinov
Oral presentation, NeurIPS 2020, arXiv:2002.11089

Dynamical Distance Learning for Unsupervised and Semi-Supervised Skill Discovery
Kristian Hartikainen, **Xinyang Geng**, Tuomas Haarnoja*, Sergey Levine*
ICLR 2020. arXiv:1907.08225

Improved Generalization with Curvature Regularization
Xinyang Geng, Lechao Xiao, Hossein Mobahi, Jeffrey Pennington
ICML 2018 Workshop: Modern Trends in Nonconvex Optimization for Machine Learning Workshop

Automatic Goal Generation for Reinforcement Learning Agents
Carlos Florensa*, David Held*, **Xinyang Geng***, Pieter Abbeel
ICML 2018. arXiv:1705.06366

Real-Time User-Guided Image Colorization with Learned Deep Priors
Richard Zhang*, Jun-Yan Zhu*, Phillip Isola, **Xinyang Geng**, Angela S. Lin, Tianhe Yu, Alexei A. Efros
SIGGRAPH 2017. arXiv:1705.02999

Deep Reinforcement Learning for Tensegrity Robot Locomotion
Marvin Zhang*, **Xinyang Geng***, Jonathan Bruce*, Ken Caluwaerts, Massimo Vespignani, Vytas SunSpiral, Pieter Abbeel, Sergey Levine
ICRA 2017. arXiv:1609.09049

RESEARCH EXPERIENCE

UC Berkeley — Aug 2018 – Now
Graduate Student Researcher — Berkeley, CA

- Advised by Professor Sergey Levine.
- Working on deep reinforcement learning, model-based optimization, and language models.

Google Brain — May 2022 – October 2022
Research Intern — Mountain View, CA

- Worked under the supervision of Igor Mordatch
- Worked on research projects in large language models, vision-language models and deep reinforcement learning.

Google Brain — Jul 2017 – Jul 2018
Google AI Resident — Mountain View, CA

- Worked under the supervision of Sergey Levine, Jeffrey Pennington, Hossein Mobahi, Nicolas Le Roux and Ofir Nachum.
- Worked on research projects in neural network architecture improvement, regularization techniques for neural networks and meta-reinforcement learning.

UC Berkeley AI Research Lab — Oct 2015 – May 2017
Undergraduate Research Assistant — Berkeley, CA

- Worked under the supervision of Professor Pieter Abbeel, Professor Sergey Levine and Professor Alexei Efros.
- Performed research on deep reinforcement learning for robotics, curriculum for goal conditioned reinforcement learning and image colorization.

ENGINEERING EXPERIENCE

Google — May – Aug 2015
Software Engineering Intern — Mountain View, CA

- Worked in advertisement data infrastructure engineering productivity team.
- Designed and built dashboard website for a log processing pipeline testing framework.
- Created performance visualization and alert service.

IBM — May – Aug 2014
Software Engineering Intern — Emeryville, CA

- Worked on the IBM Endpoint Manager product.
- Designed software for multiple platforms including Windows with Microsoft SQL Server and Linux with IBM DB2.
- Developed GUI in Microsoft Visual Studio with MFC.

RESEARCH SOFTWARE

OpenLLaMA: An Open Reproduction of LLaMA
GitHub: `https://github.com/openlm-research/open_llama`

We release a permissively licensed open source reproduction of Meta AI's LLaMA large language model. We provide PyTorch and JAX weights of our pre-trained OpenLLaMA 7B model, as well as evaluation results and comparison against the original LLaMA models. The OpenLLaMA model weights can serve as a drop in replacement for the original LLaMA in downstream applications.

Koala: A Dialogue Model for Academic Research
Blog post: `https://bair.berkeley.edu/blog/2023/04/03/koala`
Online demo: `https://koala.lmsys.org/`

Fine-tuned on top of Meta's LLaMA, Koala is a dialogue language model trained on dialogue data gathered from the web. Our user study shows that Koala can effectively respond to a variety of user queries, generating responses that are often preferred over Alpaca, and at least tied with ChatGPT in over half of the cases.

EasyLM
GitHub: `https://github.com/young-geng/EasyLM`

Large language models (LLMs) made easy, EasyLM is a one stop solution for pre-training, fine-tuning, evaluating and serving LLMs in JAX/Flax. EasyLM can scale up LLM training to hundreds of TPU/GPU accelerators by leveraging JAX's pjit compilation.

MLXU
GitHub: `https://github.com/young-geng/mlxu`

Machine Learning eXperiment Utilities: convenient utilities for running machine learning experiments, parsing experiment configurations and logging results.

JaxCQL
GitHub: `https://github.com/young-geng/JaxCQL`

A simple and modular implementation of the conservative Q learning (CQL) and soft actor critic (SAC) algorithm in JAX and Flax. This implementation is highly efficient, so that you can train an SAC agent in 30 minutes and a CQL agent in an hour.

RESEARCH MENTORING

Russell Mendonca (2019-2020)
Undergraduate at UC Berkeley → PhD student at CMU.
CRA Undergraduate Research Award finalists.

Brandon Trabucco (2020-2021)
Undergraduate at UC Berkeley → PhD student at CMU.

Kevin Li (2022)
5th year MS at UC Berkeley.

Sathvik Kolli (2022-2023)
5th year MS at UC Berkeley, to graduate in 2023.

Han Qi (2022-2023)
Undergraduate at UC Berkeley, to graduate in 2023.
CRA Undergraduate Research Award nomination.

SERVICES

Reviwer: ICML 2019, NeurIPS 2019, ICLR 2020, NeurIPS 2020, ICLR 2021, ICML 2021, NeurIPS 2021, Neurips 2022, ICLR 2022, ICLR 2023, ICML 2023

TECHNICAL SKILLS

Deep Learning Frameworks: JAX, PyTorch
Github Profile: `https://github.com/young-geng`